



North Sydney Girls High School

2026

HSC TRIAL EXAMINATION

Mathematics Extension 1

General Instructions

- Reading Time – 10 minutes
- Working Time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks:
70

Section I – 10 marks (pages 2 – 7)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 8 – 15)

- Attempt Questions 11 – 14
- Allow about 1 hour and 45 minutes for this section

NAME: _____

TEACHER: _____

STUDENT NUMBER:

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Question	1-10	11	12	13	14	Total
Mark	/10	/15	/15	/15	/15	/70

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple choice answer sheet for Questions 1-10.

1 Given that $\overrightarrow{OA} = \begin{pmatrix} 2 \\ -7 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$, what is $\overrightarrow{AB}$?

A. $\begin{pmatrix} 5 \\ 11 \end{pmatrix}$

B. $\begin{pmatrix} -5 \\ 11 \end{pmatrix}$

C. $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$

D. $\begin{pmatrix} -1 \\ -3 \end{pmatrix}$

2 The polynomial $p(x) = x^3 - 6x^2 + 32$ has one repeated zero of multiplicity 2 at $x = \alpha$.
What is the value of α ?

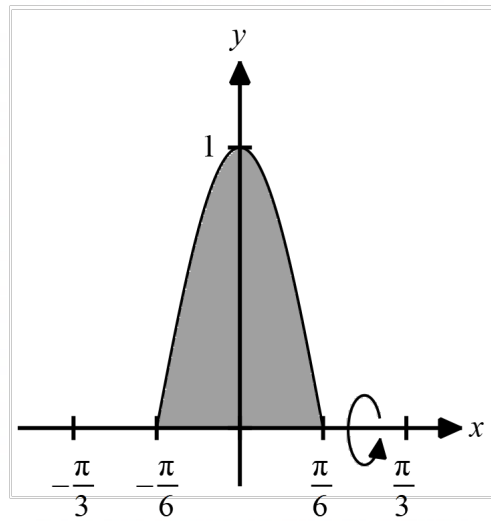
A. -2

B. 0

C. 2

D. 4

- 3 A solid of revolution is to be formed by rotating the region bounded by the x -axis and the curve $y = \cos 3x$ between $x = -\frac{\pi}{6}$ and $x = \frac{\pi}{6}$ about the x -axis, as shown in the diagram.

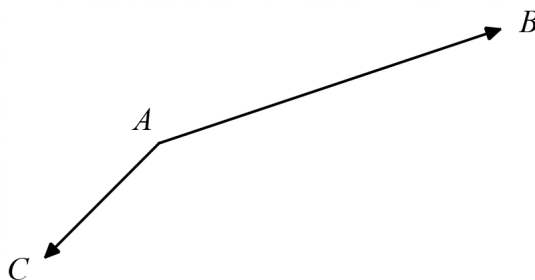


Which of the following gives the volume of solid of revolution formed?

- A. $\pi \int_0^{\frac{\pi}{6}} 1 + \cos 6x \, dx$
- B. $\pi \int_0^{\frac{\pi}{6}} \cos^2 3x \, dx$
- C. $\frac{\pi}{2} \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \cos^2 3x \, dx$
- D. $\frac{\pi}{2} \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} 1 - \cos 6x \, dx$

- 4 The projection of $\underline{u}$ onto $\underline{v}$ is given by $\left(\frac{\underline{u} \cdot \underline{v}}{|\underline{v}|^2} \right) \underline{v}$.

The diagram below shows points A , B and C , where $\overrightarrow{AB} = 3\hat{i} + \hat{j}$, $\overrightarrow{AC} = -\hat{i} - \hat{j}$.



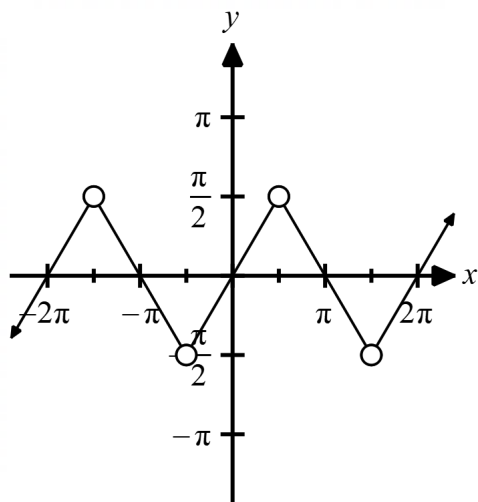
Which of the following is the projection of $\overrightarrow{AB}$ onto $\overrightarrow{AC}$?

- A. $\frac{2}{5}(3\hat{i} + \hat{j})$
- B. $\frac{-2}{5}(3\hat{i} + \hat{j})$
- C. $2(\hat{i} + \hat{j})$
- D. $2(-\hat{i} - \hat{j})$

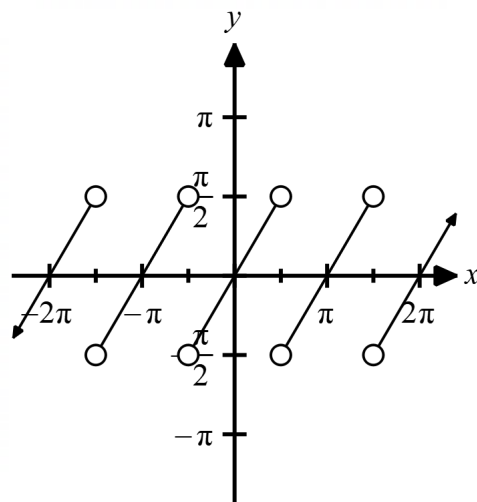
- 5 Five different schools send two prefects each to attend a conference. The group of 10 is to be seated at a circular table. In how many ways can the prefects be seated if prefects from the same school are to be seated together?
- A. $4!$
- B. $2 \times 4!$
- C. $2^4 \times 4!$
- D. $2^5 \times 4!$
- 6 You are given that $f(x)$ is a one-to-one function defined for $x \in (-\infty, \infty)$. Which of the following functions cannot have an inverse?
- A. $f(|x|)$
- B. $|f(x)|$
- C. $\frac{1}{f(x)}$
- D. $\sqrt{f(x)}$
- 7 How many distinct solutions are there to the equation $\sin x + \sin 3x = \sin 2x$ for $0 \leq x \leq 2\pi$?
- A. 5
- B. 6
- C. 7
- D. 8

8 Which graph represents the function $y = \tan^{-1}(\tan x)$?

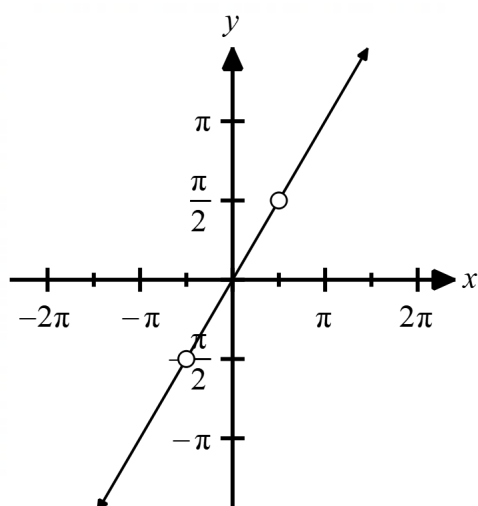
A.



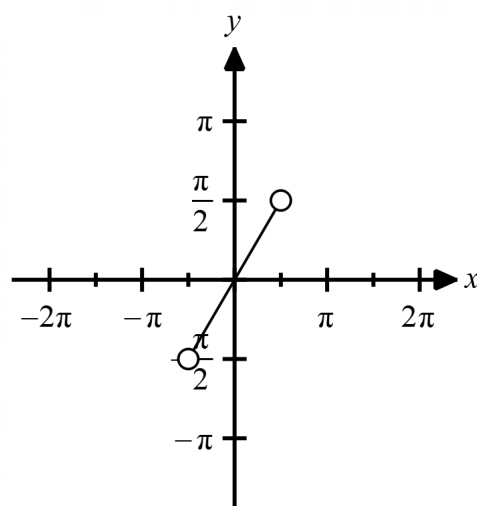
B.



C.



D.



- 9 It is known that the function $f(x)$ is one-to-one and continuous for all real x .

Let $g(x) = f^{-1}(x)$.

Which of the following statements MUST be true for any value of x where $f'(x)$ and $g'(x)$ are both defined?

A. $f'(x) = \frac{-1}{g'(x)}$

B. $f'(x) = \frac{1}{g'(x)}$

C. $f'(x)g'(x) < 0$

D. $f'(x)g'(x) > 0$

- 10 Which statement is always true for real numbers a and b where $0 \leq a < b \leq 1$?

A. $\cos^{-1} a < \cos^{-1} b$

B. $\cos^{-1} a - \sin^{-1} a > \cos^{-1} b - \sin^{-1} b$

C. $\cos^2 a < \cos^2 b$

D. $\sin a \cos a > \sin b \cos b$

End of Section I

Section II

60 marks

Attempt Questions 11-14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet

- (a) The parametric equations of a line are given below. **2**

$$x = 2 - 5t$$

$$y = 3t + 1$$

Find the Cartesian equation of this line in the form $y = mx + c$.

- (b) Find the inverse function, $f^{-1}(x)$, of the function $f(x) = e^{2x} - 1$. **2**

- (c) The graph of $y = f(x)$ is given on the separate Response Sheet for Question 11 (c). **2**

On the graph provided in the response sheet, sketch the graph of $y = \sqrt{f(x)}$, clearly showing all important features.

- (d) Consider the polynomial **3**

$$P(x) = x^3 - 2x^2 - 7x - 4.$$

Show that $x + 1$ is a factor of $P(x)$ and hence find $P(x)$ in fully factorised form.

Question 11 continues on page 9

Question 11 (continued)

(e) Solve $\frac{2x}{3-x} \leq 4$. **3**

(f) Find the coefficient of x^{10} in the expansion of $\left(\frac{2}{x} - x^3\right)^6$. **3**

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet

(a) Use Mathematical induction to prove that $4n^3 - n$ is divisible by 3 for all integers $n \geq 1$. **3**

(b) (i) Express $3 \sin x - \sqrt{3} \cos x$ in the form $2\sqrt{3} \sin(x + \alpha)$, where $0 < \alpha < 2\pi$. **2**

(ii) Hence, or otherwise, solve $3 \sin x = \sqrt{3} \cos x - \sqrt{3}$, where $0 \leq x < 2\pi$. **2**

(c) Evaluate $\int_0^{\frac{\sqrt{3}}{2}} \frac{8x^2}{\sqrt{1-x^2}} dx$ using the substitution $x = \sin \theta$. **4**

(d) Charles is a member of an elite racing group of 11 people. A team of 5 people is to be formed from the group. Use a combinatorial argument in this context to prove that **2**

$$\binom{11}{5} = \binom{10}{4} + \binom{10}{5}.$$

(e) By sketching a graph, or otherwise, solve $(|x| - 1)(|x| - 2) = 6$. **2**

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet

(a) Let $f(x) = \tan^{-1} x - \tan^{-1} \left(\frac{x-1}{x+1} \right)$, for $x > -1$.

(i) Show that $f(x) = k$ where k is a real number. 2

(ii) Hence, find the value of k . 1

(b) A particle is moving on the curve with equation 4

$$y = 2 \arcsin 3x, \quad -\frac{1}{3} \leq x \leq \frac{1}{3}.$$

The particle has coordinates (x, y) at time t , where distances are measured in metres and time in seconds.

At $t = T$, the y coordinate of the particle is $\frac{1}{3}\pi$ m and the y coordinate is increasing at a rate of 2 m/s.

Find the exact rate at which the x coordinate of the particle is changing at $t = T$.

(c) At a particular high school, 17 students, including Xavi and Yuri, are enrolled to study Mathematics Extension 1. There are three Mathematics Extension 1 classes with a maximum class size of 6. Each class is taught by a different teacher.

(i) Will there always be a class that is full? Justify your answer. 1

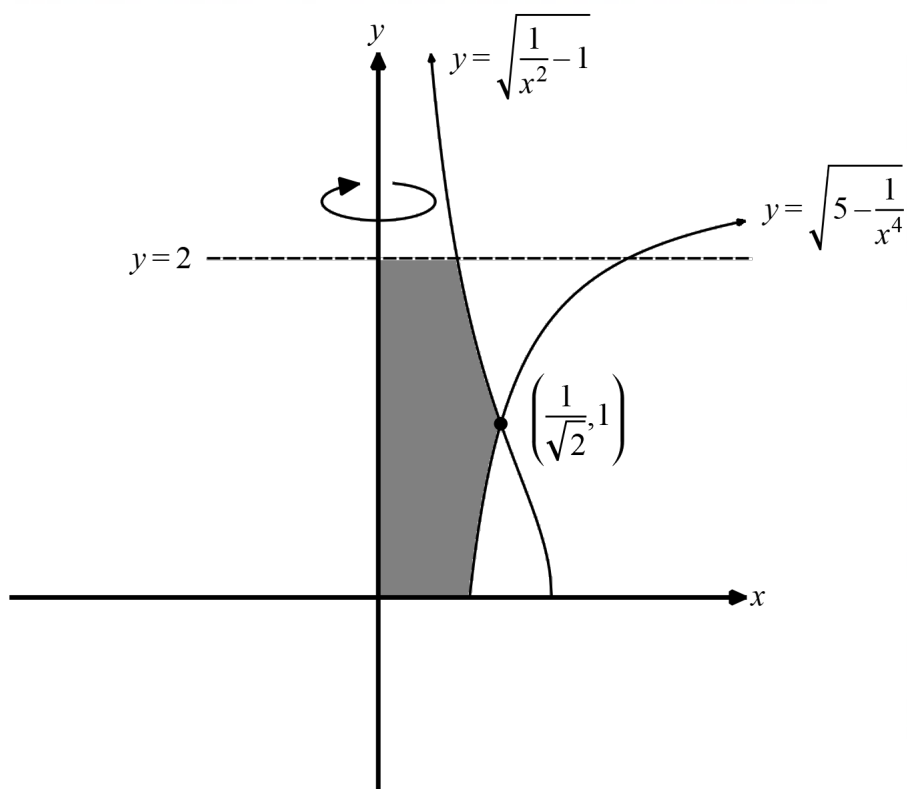
(ii) Students are randomly assigned into classes. Find the number of ways the classes can be assigned to teachers such that Xavi and Yuri are in the same class. 2

Question 13 continues on page 12

Question 13 (continued)

- (d) A solid trophy is to be made by rotating the shaded region below around the y -axis.

5



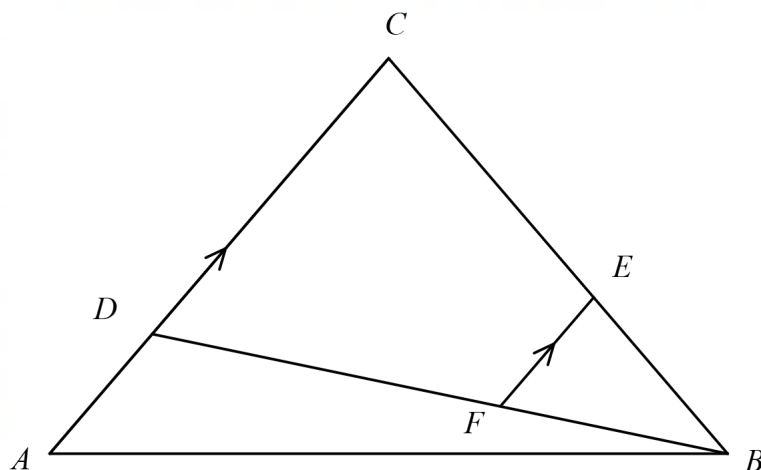
Show that the volume of the trophy is $\frac{\pi^2}{4} u^3$.

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet

- (a) In triangle ABC , D lies on AC such that $\overrightarrow{AD} = \frac{1}{4}\overrightarrow{AC}$, E lies on BC such that $\overrightarrow{BE} = \frac{1}{3}\overrightarrow{BC}$.

F lies on BD such that $FE \parallel AC$.



- (i) Show that $\overrightarrow{FE} = \overrightarrow{FB} + \frac{1}{3}\overrightarrow{BC}$. **1**
- (ii) Let $\overrightarrow{FE} = k\overrightarrow{DC}$. Use vector methods to show that $\overrightarrow{FE} = \overrightarrow{AD}$. **3**

Question 14 continues on page 14

Question 14 (continued)

- (b) The polynomial $p(x) = x^3 + bx^2 + cx + d$ has stationary points at $x = \alpha$ and $x = \beta$, where $\alpha \neq \beta$.

(i) Show that $b^2 > 3c$. **2**

(ii) You are given that $m^3 - n^3 = (m - n)(m^2 + mn + n^2)$. (Do NOT prove this) **3**

Show that
$$\frac{p(\alpha) - p(\beta)}{\alpha - \beta} = \frac{2}{9}(3c - b^2).$$

(iii) Hence, or otherwise, find the relationship between α and β such that $p(\alpha) > p(\beta)$. **1**

Question 14 continues on page 15

Question 14 (continued)

- (c) Particle A is projected from the origin with initial speed $V \text{ m s}^{-1}$ at an angle θ with the horizontal plane. The position vector of particle A , t seconds after it is projected, is given by

$$\vec{r}_A(t) = \begin{pmatrix} Vt \cos \theta \\ Vt \sin \theta - \frac{1}{2}gt^2 \end{pmatrix}. \quad (\text{Do NOT prove this})$$

- (i) Show that the velocity vector of particle A is given by $\vec{v}_A(t) = \begin{pmatrix} V \cos \theta \\ V \sin \theta - gt \end{pmatrix}$. 1

After T seconds, particle B is projected from the origin with the same initial speed and angle as particle A . The position vector of particle B is given by

$$\vec{r}_B(t) = \begin{pmatrix} V(t-T) \cos \theta \\ V(t-T) \sin \theta - \frac{1}{2}g(t-T)^2 \end{pmatrix}, \quad t \geq T. \quad (\text{Do NOT prove this})$$

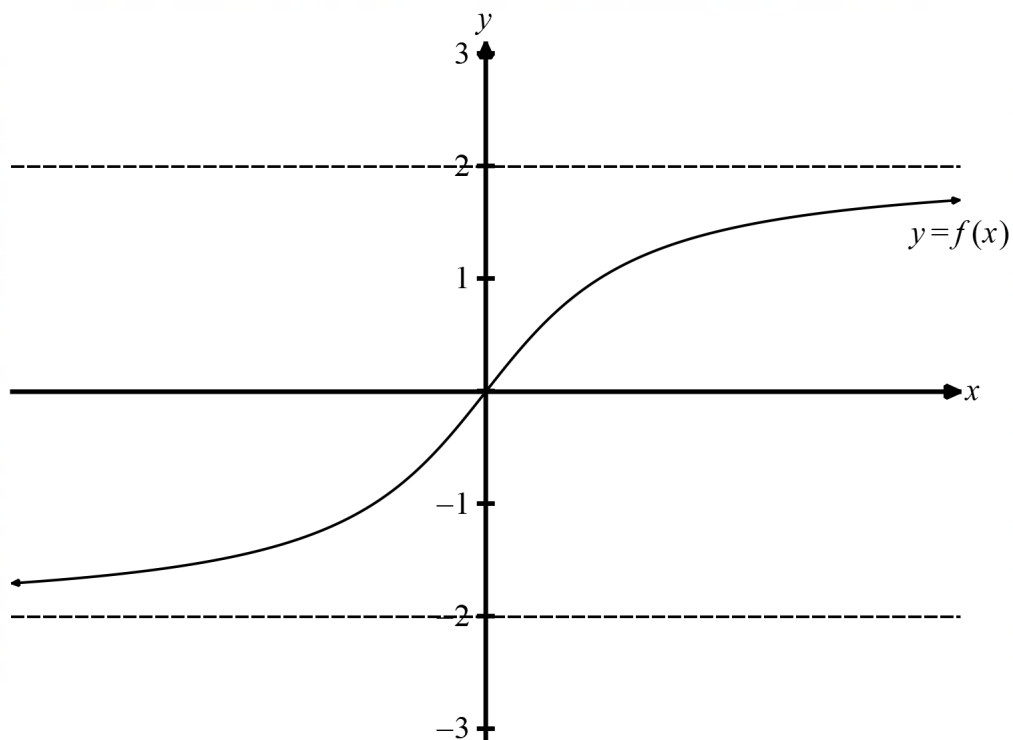
- (ii) Show that the particles will never travel in perpendicular directions at any time t if $T < \frac{2V \cos \theta}{g}$. 4

End of paper

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Question 11 (c) – Response Sheet

11 (c) The graph of $y = f(x)$ is shown below.



On the same grid above, sketch the graph of $y = \sqrt{f(x)}$, clearly showing all important features.

2

Place this sheet inside your Question 11 answer booklet.



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- Attempt Questions 1 – 10
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Section II – 60 marks (pages 8 – 15)

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Question	1-10	11	12	13	14	Total
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Section I

10 marks

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A. $\begin{pmatrix} 5 \\ 11 \end{pmatrix}$

B. $\begin{pmatrix} -5 \\ 11 \end{pmatrix}$

C. $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$

D. $\begin{pmatrix} -1 \\ -3 \end{pmatrix}$

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} -3 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ -7 \end{pmatrix} = \begin{pmatrix} -5 \\ 11 \end{pmatrix}$$

2 The polynomial $p(x) = x^3 - 6x^2 + 32$ has one repeated zero of multiplicity 2 at $x = \alpha$. What is the value of α ?

A. -2

B. 0

C. 2

D. 4

$$p(\alpha) = 0 \text{ and } p'(\alpha) = 0$$

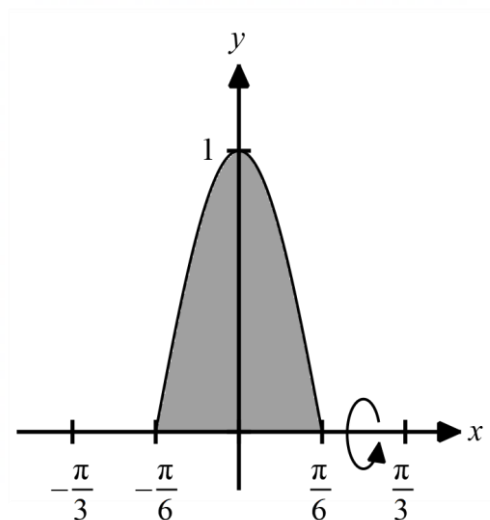
$$p'(x) = 3x^2 - 12x = 3x(x - 4)$$

$$\therefore p'(4) = 0 \text{ and } p'(0) = 0$$

$$p(4) = 4^3 - 6(4)^2 + 32 = 64 - 96 + 32 = 0$$

$$p(0) = 32 \neq 0$$

- 3 A solid of revolution is to be formed by rotating the region bounded by the x -axis and the curve $y = \cos 3x$ between $x = -\frac{\pi}{6}$ and $x = \frac{\pi}{6}$ about the x -axis, as shown in the diagram.



Which of the following gives the volume of solid of revolution formed?

A. $\pi \int_0^{\frac{\pi}{6}} 1 + \cos 6x \, dx$

B. $\pi \int_0^{\frac{\pi}{6}} \cos^2 3x \, dx$

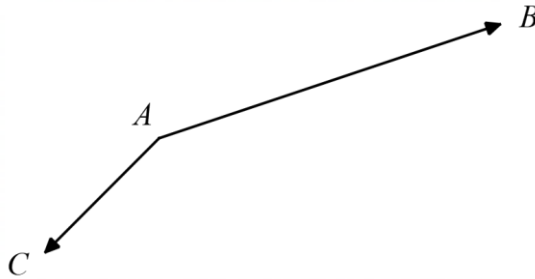
C. $\frac{\pi}{2} \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \cos^2 3x \, dx$

D. $\frac{\pi}{2} \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} 1 - \cos 6x \, dx$

$$\begin{aligned}
 V &= \pi \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \cos^2 3x \, dx = 2\pi \int_0^{\frac{\pi}{6}} \cos^2 3x \, dx \quad (\because y = \cos^2 3x \text{ is even}) \\
 &= 2\pi \int_0^{\frac{\pi}{6}} \frac{1}{2} (1 + \cos 6x) \, dx \\
 &= \pi \int_0^{\frac{\pi}{6}} 1 + \cos 6x \, dx
 \end{aligned}$$

- 4 The projection of $\underline{u}$ onto $\underline{v}$ is given by $\left(\frac{\underline{u} \cdot \underline{v}}{|\underline{v}|^2} \right) \underline{v}$.

The diagram below shows points A , B and C , where $\overrightarrow{AB} = 3\hat{i} + \hat{j}$, $\overrightarrow{AC} = -\hat{i} - \hat{j}$.



Which of the following is the projection of $\overrightarrow{AB}$ onto $\overrightarrow{AC}$?

- A. $\frac{2}{5}(3\hat{i} + \hat{j})$
- B. $\frac{-2}{5}(3\hat{i} + \hat{j})$
- C. $2(\hat{i} + \hat{j})$
- D. $2(-\hat{i} - \hat{j})$

Quick check: projection vector is parallel to $\overrightarrow{AC}$ but in opposite direction to $\overrightarrow{AC}$. Only C can be correct.

$$\left(\frac{\underline{u} \cdot \underline{v}}{|\underline{v}|^2} \right) \underline{v} = \left(\frac{3 \times (-1) + 1 \times (-1)}{\left(\sqrt{(-1)^2 + (-1)^2} \right)^2} \right) \overrightarrow{AC} = \frac{-4}{2}(-\hat{i} - \hat{j}) = 2(\hat{i} + \hat{j})$$

- 5 Five different schools send two prefects each to attend a conference. The group of 10 is to be seated at a circular table. In how many ways can the prefects be seated if prefects from the same school are to be seated together?

- A. $4!$
- B. $2 \times 4!$
- C. $2^4 \times 4!$
- D. $2^5 \times 4!$

Group the pairs of prefects from the same school. There are five groups to arrange in a circular table in $4!$ ways. Each pair within the group can be arranged in $2! = 2$ ways for a total of 2^5 ways. Therefore, total number of arrangements is $2^5 \times 4!$.

- 6 You are given that $f(x)$ is a one-to-one function defined for $x \in (-\infty, \infty)$. Which of the following functions cannot have an inverse?

- A. $f(|x|)$
- B. $|f(x)|$
- C. $\frac{1}{f(x)}$
- D. $\sqrt{f(x)}$

$f(|x|)$ is always even. Therefore, it cannot have an inverse (its inverse will not be a function)

7 How many distinct solutions are there to the equation $\sin x + \sin 3x = \sin 2x$ for $0 \leq x \leq 2\pi$?

A. 5

B. 6

C. 7

D. 8

Using sum-to-product result

$$\frac{1}{2}(\sin x + \sin 3x) = \frac{1}{2}(\sin(2x + x) + \sin(2x - x)) = \sin 2x \cos x$$

$$\sin x + \sin 3x = \sin 2x$$

$$2 \sin 2x \cos x = \sin 2x$$

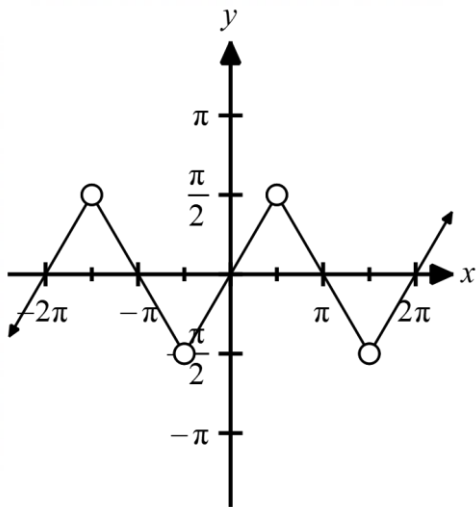
$$\sin 2x(2 \cos x - 1) = 0$$

$$\sin 2x = 0 \quad \cos x = \frac{1}{2}$$

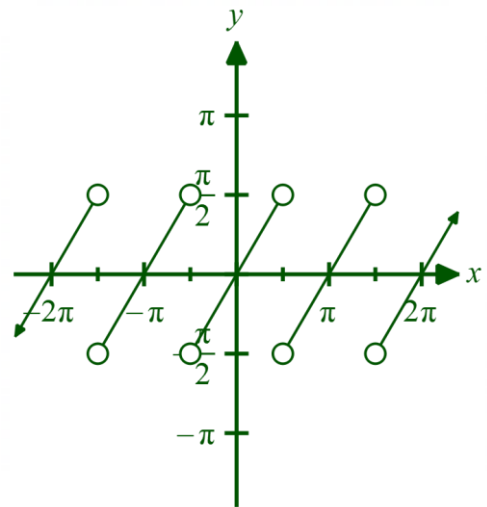
$\sin 2x = 0$ has 5 solutions for $0 \leq x \leq 2\pi$ and $\cos x = \frac{1}{2}$ has 2 solutions for $0 \leq x \leq 2\pi$, and the solutions are unique. Therefore, there are 7 solutions in total.

8 Which graph represents the function $y = \tan^{-1}(\tan x)$?

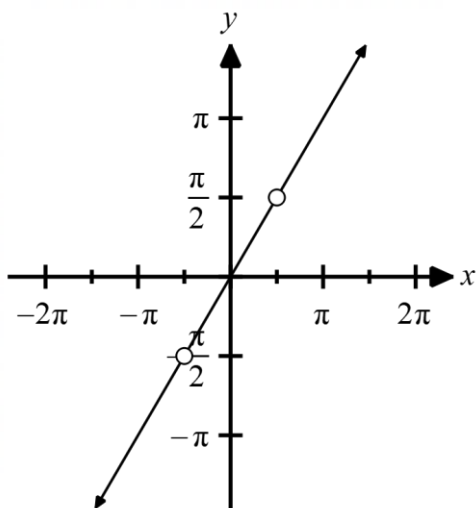
A.



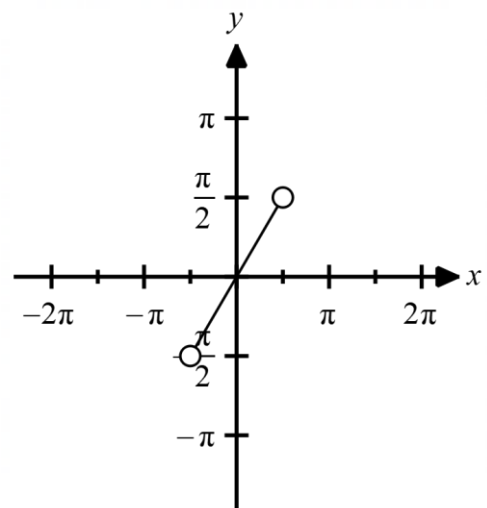
B.



C.



D.



- 9 It is known that the function $f(x)$ is one-to-one and continuous for all real x .

Let $g(x) = f^{-1}(x)$.

Which of the following statements MUST be true for any value of x where $f'(x)$ and $g'(x)$ are both defined?

- A. $f'(x) = \frac{-1}{g'(x)}$
- B. $f'(x) = \frac{1}{g'(x)}$
- C. $f'(x)g'(x) < 0$
- D. $f'(x)g'(x) > 0$

$f(x)$ and $g(x)$ are either both monotonically increasing or both monotonically decreasing.

Therefore:

- $f'(x) > 0$ and $g'(x) > 0$
- $f'(x) < 0$ and $g'(x) < 0$

Both result in $f'(x)g'(x) > 0$

- 10 Which statement is always true for real numbers a and b where $0 \leq a < b \leq 1$?

- A. $\cos^{-1} a < \cos^{-1} b$
- B. $\cos^{-1} a - \sin^{-1} a > \cos^{-1} b - \sin^{-1} b$
- C. $\cos^2 a < \cos^2 b$
- D. $\sin a \cos a > \sin b \cos b$

Consider

$$f(x) = \cos^{-1} x - \sin^{-1} x = \cos^{-1} x - \left(\frac{\pi}{2} - \cos^{-1} x \right) = 2 \cos^{-1} x - \frac{\pi}{2}$$

$f(x)$ is a decreasing function for $0 \leq x \leq 1$ as $2 \cos^{-1} x$ is a decreasing function.

$$\therefore \cos^{-1} a - \sin^{-1} a > \cos^{-1} b - \sin^{-1} b$$

Section II

60 marks

Attempt Questions 11-14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet

- (a) The parametric equations of a line are given below.

2

$$x = 2 - 5t$$

$$y = 3t + 1$$

Find the Cartesian equation of this line in the form $y = mx + c$.

$$x = 2 - 5t$$

$$t = \frac{2 - x}{5}$$

$$y = 3t + 1$$

$$y = 3\left(\frac{2 - x}{5}\right) + 1$$

$$y = \frac{-3x}{5} + \frac{6}{5} + 1$$

$$y = \frac{-3x}{5} + \frac{11}{5}$$

- (b) Find the inverse function, $f^{-1}(x)$, of the function $f(x) = e^{2x} - 1$.

2

$$x = e^{2y} - 1$$

$$e^{2y} = x + 1$$

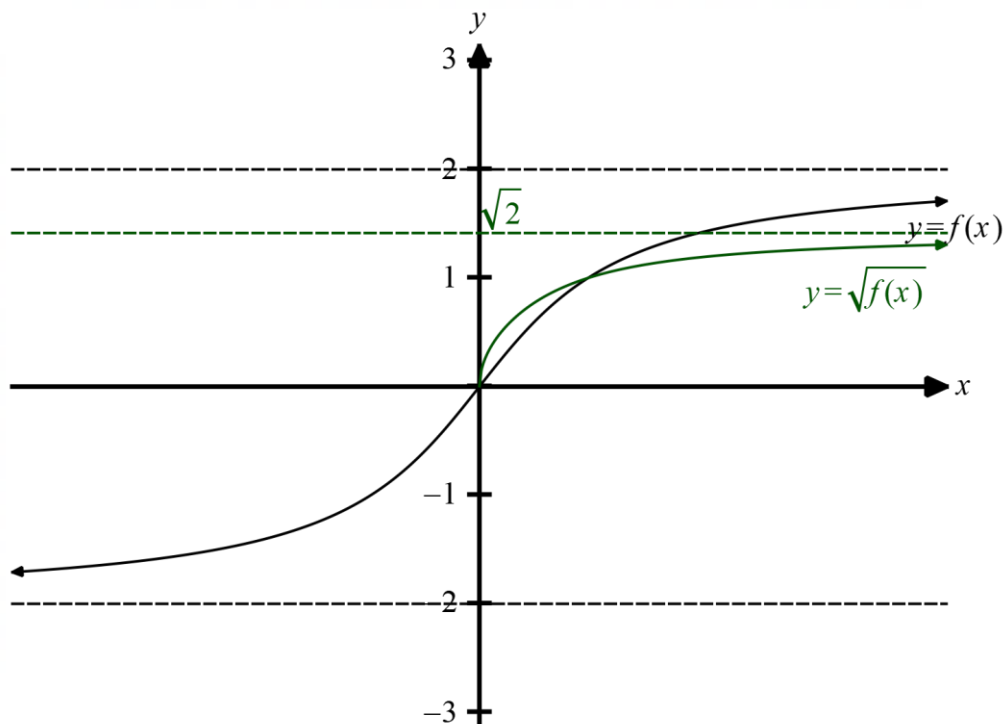
$$2y = \ln(x + 1)$$

$$y = \frac{1}{2} \ln(x + 1)$$

$$f^{-1}(x) = \frac{1}{2} \ln(x + 1)$$

- (c) The graph of $y = f(x)$ is given on the separate Response Sheet for Question 11 (c). 2

On the graph provided in the response sheet, sketch the graph of $y = \sqrt{f(x)}$, clearly showing all important features.



- (d) Consider the polynomial 3

$$P(x) = x^3 - 2x^2 - 7x - 4.$$

Show that $x + 1$ is a factor of $P(x)$ and hence find $P(x)$ in fully factorised form.

$$P(-1) = (-1)^3 - 2(-1)^2 - 7(-1) - 4 = -1 - 2 + 7 - 4 = 0$$

$$\begin{array}{r} x^2 - 3x - 4 \\ (x+1) \overline{) x^3 - 2x^2 - 7x - 4} \\ \underline{x^3 + x^2} \\ -3x^2 - 7x - 4 \\ \underline{-3x^2 - 3x} \\ -4x - 4 \\ \underline{-4x - 4} \end{array}$$

$$P(x) = (x+1)(x^2 - 3x - 4) = (x+1)(x+1)(x-4) = (x+1)^2(x-4)$$

Question 11 (continued)

(e) Solve $\frac{2x}{3-x} \leq 4$.

3

$$\frac{2x}{3-x} \times (3-x)^2 \leq 4 \times (3-x)^2$$

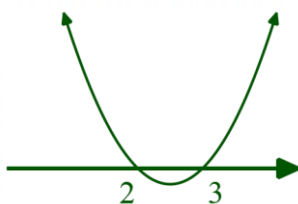
$$2x(3-x) \leq 4(3-x)^2$$

$$4(3-x)^2 - 2x(3-x) \geq 0$$

$$(3-x)[4(3-x) - 2x] \geq 0$$

$$(3-x)(12-6x) \geq 0$$

$$6(x-3)(x-2) \geq 0$$



$$\therefore x \leq 2, x \geq 3 \quad (\because x \neq 3)$$

(f) Find the coefficient of x^{10} in the expansion of $\left(\frac{2}{x} - x^3\right)^6$.

3

The expansion will contain terms in the form of:

$$\begin{aligned} & \binom{6}{r} \left(\frac{2}{x}\right)^{6-r} (-x^3)^r \\ &= \binom{6}{r} 2^{6-r} x^{r-6} (-1)^r x^{3r} \\ &= \binom{6}{r} 2^{6-r} (-1)^r x^{4r-6} \end{aligned}$$

To obtain x^{10} :

$$4r - 6 = 10$$

$$r = 4$$

coefficient of x^{10} :

$$\binom{6}{4} 2^{6-4} (-1)^4 = 60$$

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet

- (a) Use Mathematical induction to prove that $4n^3 - n$ is divisible by 3 for all integers $n \geq 1$. **3**

When $n = 1$

$$\begin{aligned}4(1)^3 - 1 &= 3 = 3 \times 1 \\ \therefore \text{statement is true for } n &= 1\end{aligned}$$

Assume the statement is true for $n = k$

$$4k^3 - k = 3A \text{ where } A \in \mathbb{Z}$$

When $n = k + 1$

$$\begin{aligned}&4(k+1)^3 - (k+1) \\&= 4(k^3 + 3k^2 + 3k + 1) - k - 1 \\&= (4k^3 - k) + 3 \times 4k^2 + 3 \times 4k + 4 - 1 \\&= 3A + 3(4k^2 + 4k + 1) \\&= 3(A + 4k^2 + 4k + 1) \\&= 3B \text{ (where } B = A + 4k^2 + 4k + 1 \in \mathbb{Z}, \because A, k \in \mathbb{Z}) \\&\therefore \text{statement is true for } n = k + 1 \text{ if it is true for } n = k.\end{aligned}$$

Hence the statement is true for all integers $n \geq 1$ by Mathematical Induction.

- (b) (i) Express $3\sin x - \sqrt{3}\cos x$ in the form $2\sqrt{3}\sin(x+\alpha)$, where $0 < \alpha < 2\pi$.

2

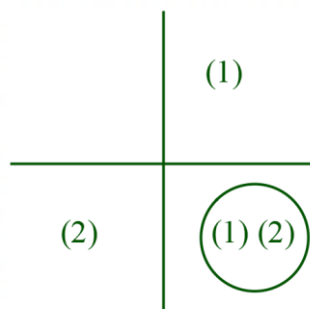
$$3\sin x - \sqrt{3}\cos x = 2\sqrt{3}(\sin x \cos \alpha + \cos x \sin \alpha)$$

$$3\sin x - \sqrt{3}\cos x = (2\sqrt{3}\cos \alpha)\sin x + (2\sqrt{3}\sin \alpha)\cos x$$

By equating coefficients of $\sin x$ and $\cos x$:

$$3 = 2\sqrt{3}\cos \alpha \quad (1) \qquad -\sqrt{3} = 2\sqrt{3}\sin \alpha \quad (2)$$

$$\cos \alpha = \frac{3}{2\sqrt{3}} \quad (1) \qquad \sin \alpha = \frac{-\sqrt{3}}{2\sqrt{3}} \quad (2)$$



$\therefore \alpha$ must be in the 4th quadrant

$$\cos \alpha = \frac{3}{2\sqrt{3}}$$

$$\cos \alpha = \frac{\sqrt{3}}{2}$$

$$\alpha = \frac{11\pi}{6}$$

$$3\sin x - \sqrt{3}\cos x = 2\sqrt{3}\sin\left(x + \frac{11\pi}{6}\right)$$

- (ii) Hence, or otherwise, solve $3\sin x = \sqrt{3}\cos x - \sqrt{3}$, where $0 \leq x < 2\pi$.

2

$$3\sin x = \sqrt{3}\cos x - \sqrt{3}$$

$$3\sin x - \sqrt{3}\cos x = -\sqrt{3}$$

$$2\sqrt{3}\sin\left(x + \frac{11\pi}{6}\right) = -\sqrt{3} \quad (\text{from (i)})$$

$$\sin\left(x + \frac{11\pi}{6}\right) = \frac{-1}{2}$$

$$x + \frac{11\pi}{6} = \frac{11\pi}{6}, \frac{19\pi}{6}$$

$$x = 0, \frac{4\pi}{3}$$

- (c) Evaluate $\int_0^{\frac{\sqrt{3}}{2}} \frac{8x^2}{\sqrt{1-x^2}} dx$ using the substitution $x = \sin \theta$. 4

$$\begin{aligned} x &= \sin \theta \\ \frac{dx}{d\theta} &= \cos \theta && \text{when } x = \frac{\sqrt{3}}{2}, \theta = \frac{\pi}{3} \\ dx &= \cos \theta d\theta && \text{when } x = 0, \theta = 0 \end{aligned}$$

$$\begin{aligned} \int_0^{\frac{\sqrt{3}}{2}} \frac{8x^2}{\sqrt{1-x^2}} dx &= \int_0^{\frac{\pi}{3}} \frac{8\sin^2 \theta}{\sqrt{1-\sin^2 \theta}} \cdot \cos \theta d\theta = \int_0^{\frac{\pi}{3}} \frac{8\sin^2 \theta}{\sqrt{\cos^2 \theta}} \cdot \cos \theta d\theta = \int_0^{\frac{\pi}{3}} 8\sin^2 \theta d\theta \\ \int_0^{\frac{\pi}{3}} 8\sin^2 \theta d\theta &= 4 \int_0^{\frac{\pi}{3}} 1 - \cos 2\theta d\theta = 4 \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{3}} = 4 \left[\left(\frac{\pi}{3} - \frac{1}{2} \sin \left(2 \cdot \frac{\pi}{3} \right) \right) - \left(0 - \frac{1}{2} \sin(2 \cdot 0) \right) \right] \\ &= 4 \left(\frac{\pi}{3} - \frac{\sqrt{3}}{4} \right) \end{aligned}$$

- (d) Charles is a member of an elite racing group of 11 people. A team of 5 people is to be formed from the group. Use a combinatorial argument in this context to prove that 2

$$\binom{11}{5} = \binom{10}{4} + \binom{10}{5}.$$

LHS:

Total number of teams that can be formed by picking 5 people from 11: $\binom{11}{5}$

RHS:

Case 1: Charles is on the team

Picking the remaining 4 people on the team from the remaining 10 people: $\binom{10}{4}$

Case 2: Charles is not on the team

Picking the remaining 5 people on the team from the 10 people (excluding Charles): $\binom{10}{5}$

$$\text{Total: } \binom{10}{4} + \binom{10}{5}$$

(e) By sketching a graph, or otherwise, solve $(|x|-1)(|x|-2)=6$.

2

Let $f(x)=(x-1)(x-2)$

Solutions to $f(x)=6$,

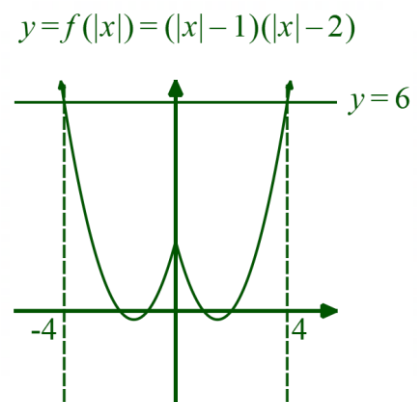
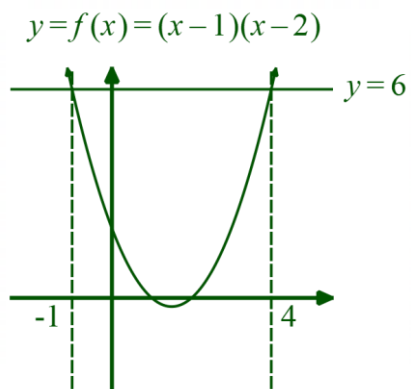
$$(x-1)(x-2)=6$$

$$x^2-3x-4=0$$

$$(x-4)(x+1)=0$$

$$x=-1, 4$$

Consider the graph of $f(x)=(x-1)(x-2)$



$$\therefore x=-4, 4$$

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet

(a) Let $f(x) = \tan^{-1} x - \tan^{-1} \left(\frac{x-1}{x+1} \right)$, for $x > -1$.

(i) Show that $f(x) = k$ where k is a real number.

2

$$\begin{aligned} f'(x) &= \frac{1}{1+x^2} - \frac{1}{1+\left(\frac{x-1}{x+1}\right)^2} \times \frac{(x+1)(1) - (x-1)(1)}{(x+1)^2} \\ &= \frac{1}{1+x^2} - \frac{2}{(x+1)^2 + (x-1)^2} \\ &= \frac{1}{1+x^2} - \frac{2}{x^2 + 2x + 1 + x^2 - 2x + 1} \\ &= \frac{1}{1+x^2} - \frac{2}{2x^2 + 2} \\ &= \frac{1}{1+x^2} - \frac{1}{1+x^2} \\ &= 0 \end{aligned}$$

$\therefore f'(x) = 0$ then $f(x)$ must be a constant. $\therefore f(x) = k$

(ii) Hence, find the value of k .

1

$$\begin{aligned} f(0) &= \tan^{-1} 0 - \tan^{-1} \left(\frac{0-1}{0+1} \right) = 0 - \tan^{-1}(-1) = 0 - \left(-\frac{\pi}{4} \right) = \frac{\pi}{4} \\ \therefore k &= \frac{\pi}{4} \end{aligned}$$

(b) A particle is moving on the curve with equation

4

$$y = 2 \arcsin 3x, \quad -\frac{1}{3} \leq x \leq \frac{1}{3}.$$

The particle has coordinates (x, y) at time t , where distances are measured in metres and time in seconds.

At $t = T$, the y coordinate of the particle is $\frac{1}{3}\pi$ m and the y coordinate is increasing at a rate of 2 m/s.

Find the exact rate at which the x coordinate of the particle is changing at $t = T$.

$$\frac{dy}{dx} = 2 \left(\frac{1}{\sqrt{1-(3x)^2}} \right) (3) = \frac{6}{\sqrt{1-9x^2}}$$

when $y = \frac{1}{3}\pi$

$$\frac{1}{3}\pi = 2 \arcsin 3x$$

$$\arcsin 3x = \frac{\pi}{6}$$

$$3x = \frac{1}{2}$$

$$x = \frac{1}{6}$$

when $x = \frac{1}{6}$

$$\frac{dy}{dx} = \frac{6}{\sqrt{1-9\left(\frac{1}{6}\right)^2}} = \frac{6}{\sqrt{\frac{3}{4}}} = \frac{12}{\sqrt{3}}$$

$$\frac{dx}{dy} = \frac{\sqrt{3}}{12}$$

At $t = T$

$$\frac{dx}{dt} = \frac{dx}{dy} \times \frac{dy}{dt} = \frac{\sqrt{3}}{12} \times 2 = \frac{\sqrt{3}}{6} \text{ m/s}$$

- (c) At a particular high school, 17 students, including Xavi and Yuri, are enrolled to study Mathematics Extension 1. There are three Mathematics Extension 1 classes with a maximum class size of 6. Each class is taught by a different teacher.

- (i) Will there always be a class that is full? Justify your answer.

1

$$\frac{\text{number of students}}{\text{number of classes}} = \frac{17}{3} = 5.66... > 5$$

∴ Using the Pigeonhole principle, at least one class has 6 students. There will always be a class that is full.

Or

Distributing 15 students evenly into the 3 classes evenly will give 5 students in each class. The 16th must be distributed to a class and make that class full. Therefore, there'll always be a class that is full

- (ii) Students are randomly assigned into classes. Find the number of ways the classes can be assigned to teachers such that Xavi and Yuri are in the same class.

2

The only possible class sizes are 2 classes of 6 students and 1 class of 5 students. Consider two cases:

Case 1: Xavi and Yuri are in a class of 6 students

$${}^{15}C_4 \times {}^{11}C_6 \times {}^5C_5 \times 3! = 3783780$$

Case 2: Xavi and Yuri are in a class of 5 students

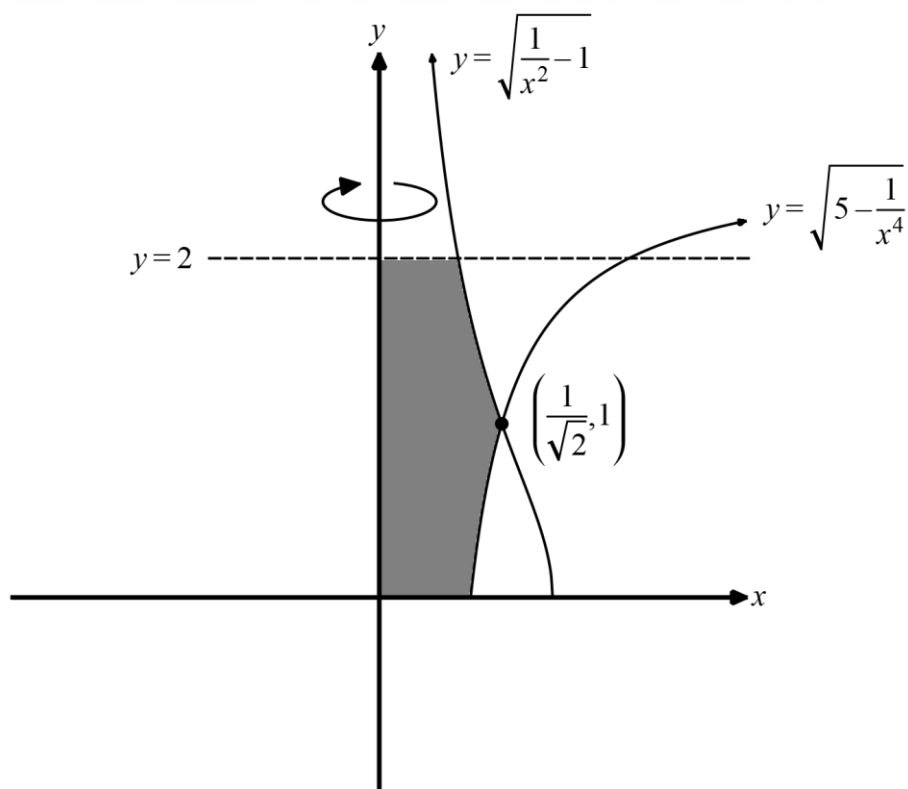
$$\frac{{}^{15}C_3 \times {}^{12}C_6 \times {}^6C_5 \times 3!}{2!} = 1261260$$

Total: 5045040

Question 13 (continued)

- (d) A solid trophy is to be made by rotating the shaded region below around the y -axis.

5



Show that the volume of the trophy is $\frac{\pi^2}{4} u^3$.

$$\begin{aligned} y &= \sqrt{5 - \frac{1}{x^4}} \\ y^2 &= 5 - \frac{1}{x^4} \\ x^4 &= \frac{1}{5 - y^2} \\ x^2 &= \frac{1}{\sqrt{5 - y^2}} \end{aligned}$$

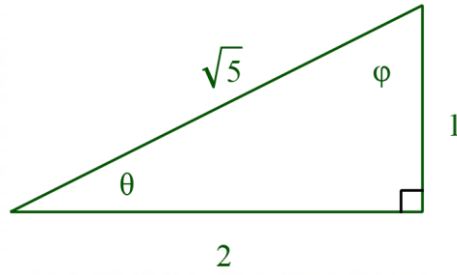
$$\begin{aligned} y &= \sqrt{\frac{1}{x^2} - 1} \\ y^2 &= \frac{1}{x^2} - 1 \\ x^2 &= \frac{1}{1 + y^2} \end{aligned}$$

$$V_1 = \pi \int_0^1 \frac{1}{\sqrt{5-y^2}} dy = \pi \left[\sin^{-1} \left(\frac{y}{\sqrt{5}} \right) \right]_0^1 = \pi \sin^{-1} \frac{1}{\sqrt{5}}$$

$$V_2 = \pi \int_1^2 \frac{1}{1+y^2} dy = \pi \left[\tan^{-1} y \right]_1^2 = \pi \left(\tan^{-1} 2 - \tan^{-1} 1 \right) = \pi \left(\tan^{-1} 2 - \frac{\pi}{4} \right)$$

$$V_{total} = \pi \left(\sin^{-1} \frac{1}{\sqrt{5}} + \tan^{-1} 2 - \frac{\pi}{4} \right)$$

Consider the following triangle



$$\theta + \phi = \frac{\pi}{2}$$

$$\sin^{-1} \frac{1}{\sqrt{5}} + \tan^{-1} \frac{2}{1} = \frac{\pi}{2}$$

$$\therefore V_{total} = \pi \left(\sin^{-1} \frac{1}{\sqrt{5}} + \tan^{-1} 2 - \frac{\pi}{4} \right) = \pi \left(\frac{\pi}{2} - \frac{\pi}{4} \right) = \frac{\pi^2}{4} u^3$$

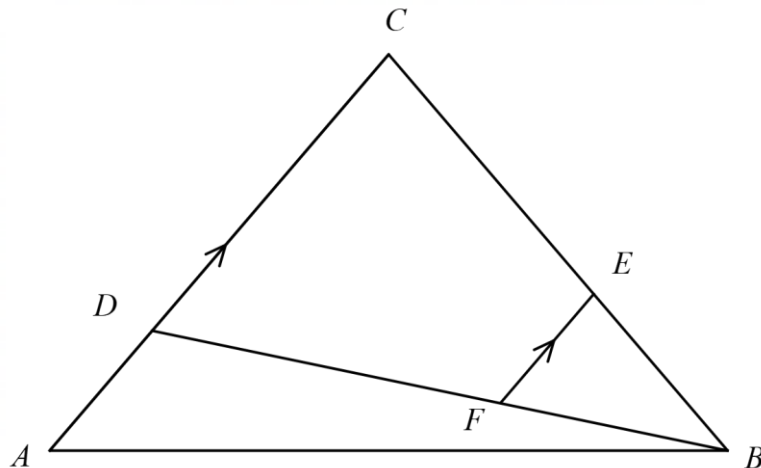
End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet

- (a) In triangle ABC , D lies on AC such that $\overrightarrow{AD} = \frac{1}{4}\overrightarrow{AC}$, E lies on BC such that

$$\overrightarrow{BE} = \frac{1}{3}\overrightarrow{BC}.$$

F lies on BD such that $FE \parallel AC$.



- (i) Show that $\overrightarrow{FE} = \overrightarrow{FB} + \frac{1}{3}\overrightarrow{BC}$.

1

$$\overrightarrow{FE} = \overrightarrow{FB} + \overrightarrow{BE} = \overrightarrow{FB} + \frac{1}{3}\overrightarrow{BC}$$

- (ii) Let $\overrightarrow{FE} = k\overrightarrow{DC}$. Use vector methods to show that $\overrightarrow{FE} = \overrightarrow{AD}$.

3

$$k\overrightarrow{DC} = \overrightarrow{FB} + \frac{1}{3}\overrightarrow{BC}$$

$$k(\overrightarrow{DB} + \overrightarrow{BC}) = \overrightarrow{FB} + \frac{1}{3}\overrightarrow{BC}$$

$$k(\lambda\overrightarrow{FB} + \overrightarrow{BC}) = \overrightarrow{FB} + \frac{1}{3}\overrightarrow{BC} \quad (\because \overrightarrow{DB} \parallel \overrightarrow{FB})$$

$$k\lambda\overrightarrow{FB} + k\overrightarrow{BC} = \overrightarrow{FB} + \frac{1}{3}\overrightarrow{BC}$$

$$\therefore k = \frac{1}{3}, \overrightarrow{FE} = \frac{1}{3}\overrightarrow{DC}, \overrightarrow{DC} = 3\overrightarrow{FE}$$

$$\overrightarrow{AD} = \frac{1}{4}(\overrightarrow{AD} + \overrightarrow{DC})$$

$$\frac{3}{4}\overrightarrow{AD} = \frac{1}{4}\overrightarrow{DC}$$

$$3\overrightarrow{AD} = \overrightarrow{DC}$$

$$3\overrightarrow{AD} = 3\overrightarrow{FE}$$

$$\overrightarrow{AD} = \overrightarrow{FE}$$

Question 14 (continued)

- (b) The polynomial $p(x) = x^3 + bx^2 + cx + d$ has stationary points at $x = \alpha$ and $x = \beta$, where $\alpha \neq \beta$.

- (i) Show that $b^2 > 3c$.

2

$$p'(x) = 3x^2 + 2bx + c$$

Since there are two stationary points, $p'(x) = 0$ must have two solutions, $\therefore \Delta > 0$

$$(2b)^2 - 4(3)(c) > 0$$

$$4b^2 - 12c > 0$$

$$4b^2 > 12c$$

$$b^2 > 3c$$

- (ii) You are given that $m^3 - n^3 = (m - n)(m^2 + mn + n^2)$. (Do NOT prove this)

3

Show that $\frac{p(\alpha) - p(\beta)}{\alpha - \beta} = \frac{2}{9}(3c - b^2)$.

$$\begin{aligned} p(\alpha) - p(\beta) &= \alpha^3 + b\alpha^2 + c\alpha + d - (\beta^3 + b\beta^2 + c\beta + d) \\ &= \alpha^3 - \beta^3 + b(\alpha^2 - \beta^2) + c(\alpha - \beta) \\ &= (\alpha - \beta)(\alpha^2 + \alpha\beta + \beta^2) + b(\alpha - \beta)(\alpha + \beta) + c(\alpha - \beta) \quad (\text{using the given result}) \\ &= (\alpha - \beta)[(\alpha^2 + \beta^2 + \alpha\beta) + b(\alpha + \beta) + c] \end{aligned}$$

$$\begin{aligned} \frac{p(\alpha) - p(\beta)}{\alpha - \beta} &= (\alpha^2 + \beta^2 + \alpha\beta) + b(\alpha + \beta) + c \\ &= (\alpha + \beta)^2 - \alpha\beta + b(\alpha + \beta) + c \\ &= \left(-\frac{2b}{3}\right)^2 - \left(\frac{c}{3}\right) + b\left(-\frac{2b}{3}\right) + c \\ &= \frac{4b^2}{9} - \frac{2b^2}{3} + \frac{2c}{3} \\ &= \frac{2c}{3} - \frac{2b^2}{9} \\ &= \frac{2}{9}(3c - b^2) \end{aligned}$$

- (iii) Hence, or otherwise, find the relationship between α and β such that $p(\alpha) > p(\beta)$.

1

Method 1: Algebraic using (i) and (ii)

From (i)

$$b^2 > 3c$$

$$3c - b^2 < 0$$

$$\therefore \frac{2}{9}(3c - b^2) < 0$$

RHS of (ii) is negative.

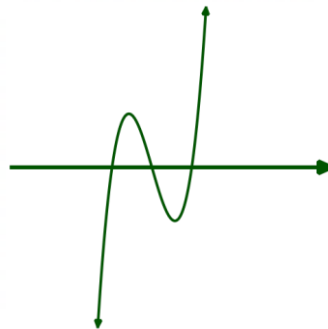
If $p(\alpha) > p(\beta)$, then $p(\alpha) - p(\beta) > 0$. For LHS of (ii) to also be negative,

$$\alpha - \beta < 0$$

$$\alpha < \beta$$

Method 2: Graphical

$p(x)$ is a monic polynomial of degree 3 with 2 stationary points, its graph must have one maxima and one minima similar to the graph below



If $p(\alpha) > p(\beta)$, then $x = \alpha$ must be the maxima and $x = \beta$ must be the minima.

$$\therefore \alpha < \beta$$

Question 14 (continued)

- (c) Particle A is projected from the origin with initial speed $V \text{ m s}^{-1}$ at an angle θ with the horizontal plane. The position vector of particle A , t seconds after it is projected, is given by

$$\vec{r}_A(t) = \begin{pmatrix} Vt \cos \theta \\ Vt \sin \theta - \frac{1}{2}gt^2 \end{pmatrix}. \quad (\text{Do NOT prove this})$$

- (i) Show that the velocity vector of particle A is given by $\vec{v}_A(t) = \begin{pmatrix} V \cos \theta \\ V \sin \theta - gt \end{pmatrix}$. 1

$$\frac{d(\vec{r}_A(t))}{dt} = \frac{d}{dt} \begin{pmatrix} Vt \cos \theta \\ Vt \sin \theta - \frac{1}{2}gt^2 \end{pmatrix}$$

$$\vec{v}_A(t) = \begin{pmatrix} V \cos \theta \\ V \sin \theta - 2\left(\frac{1}{2}gt\right) \end{pmatrix}$$

$$\vec{v}_A(t) = \begin{pmatrix} V \cos \theta \\ V \sin \theta - gt \end{pmatrix}$$

After T seconds, particle B is projected from the origin with the same initial speed and angle as particle A . The position vector of particle B is given by

$$\vec{r}_B(t) = \begin{pmatrix} V(t-T)\cos\theta \\ V(t-T)\sin\theta - \frac{1}{2}g(t-T)^2 \end{pmatrix}, \quad t \geq T. \quad (\text{Do NOT prove this})$$

- (ii) Show that the particles will never travel in perpendicular directions at any time t if $T < \frac{2V\cos\theta}{g}$.

4

$$\vec{v}_B(t) = \begin{pmatrix} V\cos\theta \\ V\sin\theta - g(t-T) \end{pmatrix}$$

If the particles are travelling at perpendicular directions, $\vec{v}_A(t) \cdot \vec{v}_B(t) = 0$.

$$\begin{aligned} (V\cos\theta)(V\cos\theta) + (V\sin\theta - gt)(V\sin\theta - g(t-T)) &= 0 \\ V^2\cos^2\theta + V^2\sin^2\theta - gV\sin\theta(t-T) - gVt\sin\theta + g^2t(t-T) &= 0 \\ V^2 - (gV\sin\theta)t + gVT\sin\theta - gVt\sin\theta + g^2t^2 - (g^2T)t &= 0 \\ g^2t^2 - (2gV\sin\theta + g^2T)t + (V^2 + gVT\sin\theta) &= 0 \end{aligned}$$

The particles will never travel in perpendicular directions if there's no solution to the above quadratic equation in t .

$$\Delta < 0$$

$$\begin{aligned} (2gV\sin\theta + g^2T)^2 - 4g^2(V^2 + gVT\sin\theta) &< 0 \\ 4g^2V^2\sin^2\theta + 4g^3VT\sin\theta + g^4T^2 - 4g^2V^2 - 4g^3VT\sin\theta &< 0 \\ g^4T^2 &< 4g^2V^2 - 4g^2V^2\sin^2\theta \\ g^4T^2 &< 4g^2V^2(1 - \sin^2\theta) \\ T^2 &< \frac{4V^2\cos^2\theta}{g^2} \\ T &< \frac{2V\cos\theta}{g} \end{aligned}$$